

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 6: Special Approvals: RVSM, NAT HLA & Contingencies

Syllabus Source: CAE Oxford (pp. 197-234) | Pass Target: >= 90% AviationExam

1. REDUCED VERTICAL SEPARATION MINIMUM (RVSM)

RVSM reduces vertical separation between aircraft from 2,000 ft to 1,000 ft between FL 290 and FL 410 inclusive, doubling airspace capacity:

- **Mandatory RVSM Equipment (Part-SPA.RVSM):** 1. Two independent primary altimetry systems (maximum difference ± 200 ft in flight); 2. One automatic altitude-control system (autopilot holding altitude within ± 65 ft); 3. One altitude-alerting system (± 300 ft alert threshold); 4. One Mode C / Mode S altitude-reporting transponder.
- **Strategic Lateral Offset Procedures (SLOP):** In oceanic airspace (NAT HLA), pilots are authorized to fly 1 NM or 2 NM RIGHT OF TRACK to disperse wake turbulence and reduce collision risk without notifying ATC!

2. OCEANIC IN-FLIGHT CONTINGENCY PROCEDURE (NAT HLA)

If unable to maintain assigned flight level or route in NAT HLA due to engine failure, depressurization, or medical emergency:

- **Step 1: Turn:** Turn 30° left or right of track (away from nearest route/traffic).
- **Step 2: Offset:** Fly to acquire and maintain a 5 NAUTICAL MILE lateral offset from the track centerline.
- **Step 3: Level Change:** Once established on the 5 NM offset, climb or descend: By 500 FT (if below FL 410); By 1,000 FT (if at or above FL 410) to avoid standard traffic levels!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): NAT HLA Oceanic In-Flight Contingency**

Emergency Descent / Engine Failure without ATC Clearance in North Atlantic HLA: Step 1: Turn on all exterior lights and squawk 7700. Step 2: Turn 30 degrees left or right off track (away from adjacent organized tracks). Step 3: Fly until establishing a 5 NM parallel offset from original track. Step 4: Once established on the 5 NM offset: - If remaining in RVSM airspace (FL 290 - FL 410): Climb or descend by 300 ft to fly an intermediate flight level (e.g. FL 353). - If descending below FL 290: Descend to suitable altitude. Step 5: Broadcast MAYDAY on 121.5 MHz and air-to-air 123.45 MHz.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Executing Oceanic Engine-Out Contingency

SCENARIO (Transatlantic Emergency Descent Drill): A twin-jet airliner is cruising at FL 370 in the North Atlantic (NAT HLA) on Track Charlie (Track 090° True): - An engine fails catastrophically! - Unable to maintain FL 370, the crew cannot establish radio communication with Shanwick ATC immediately QUESTION: What precise maneuver must the crew execute under ICAO NAT Doc 007 contingency rules? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Initial Action: - Select Maximum Continuous Thrust on good engine, set transponder to 7700, turn on all exterior lights! Step 2: Initiate Lateral Turn: - Turn 30° to the RIGHT (heading 120°) or LEFT (heading 060°) away from conflicting tracks. Step 3: Establish 5 NM Lateral Offset: - Fly out until 5.0 NM away from the centerline of Track Charlie. - Turn back onto track heading (090° True) to parallel the track 5 NM displaced. Step 4: Select Contingency Level (Divert from FL 370): - Normal traffic flies at whole flight levels: FL 370, FL 360, FL 350, FL 340. - To avoid hitting opposing or same-direction traffic below: - Aircraft must descend to an INTERMEDIATE 500 FT LEVEL! - Drift down to: FL 335, FL 315, or FL 295 (500 ft offset from normal traffic levels)! FINAL ANSWER: Turn 30° to establish 5 NM offset, then descend to a 500-ft offset flight level (e.g. FL 335 / FL 315).

[!] EXAM TRAP (AVIATIONEXAM) - The 500 ft vs 1,000 ft Contingency Offset Trap

- If below FL 410: The altitude offset is 500 FT (e.g. FL 335). - If at or above FL 410: The altitude offset is 1,000 FT. - The lateral offset is ALWAYS 5 NM in modern ICAO oceanic airspace (changed from the old 15 NM rule).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In RVSM airspace (FL 290 to FL 410), what is the maximum permissible difference between the two primary

- [A]: 50 ft
- [B]: NOT EXCEEDING 200 FEET
- [C]: 300 ft
- [D]: 500 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under EASA SPA.RVSM.110, the maximum difference between primary altimeters in level flight is 200 ft (and within 75 ft on the ground).

Q2: Under Strategic Lateral Offset Procedures (SLOP) in oceanic airspace, what lateral offsets are authorized t

- [A]: 10 NM right
- [B]: 1 NM or 2 NM RIGHT OF TRACK (or tenths of a mile up to 2 NM)
- [C]: 5 NM left
- [D]: 1 NM left

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. SLOP allows pilots to offset 1 or 2 NM right of centerline to disperse wake turbulence and minimize collision risk.

Q3: What is the lateral offset distance required when executing an oceanic in-flight contingency in the North A

- [A]: 15 NM
- [B]: Exactly 5 NAUTICAL MILES

- [C]: 30 NM
- [D]: 1 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under modern ICAO Doc 007 procedures, the oceanic contingency offset is standardized at 5 NM.